Algebra II with Trigonometry Chapter 5.5

Olympiad Problems (Gemini)

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Problems

1. **(Circumcircle Optimization)** Triangle ABC is inscribed in a circle of radius 10. If side $a = 10$ and side $b = 10\sqrt{2}$, what is the maximum possible measure of $\angle C$ in degrees?

2. **(Ambiguous Case Invariant)** In $\triangle ABC$, $\angle A = 60^\circ$, side $a = 7$, and side $b = 8$. Because this is an ambiguous case, there are two distinct valid triangles that can be formed, leading to two possible lengths for side c. What is the sum of these two lengths?

3. **(Algebraic Symmetry)** In triangle ABC, side lengths a and b are unequal, yet they satisfy the trigonometric relation $a \cos A = b \cos B$. What is the measure of $\angle C$ in degrees?

4. **(Cevian Ratios)** In $\triangle ABC$, point D lies on side BC such that $\angle BAD = 30^\circ$ and $\angle CAD = 60^\circ$. If the length of segment BD is exactly twice the length of DC, what is the exact value of the ratio $\frac{AB}{AC}$?

5. **(Elevation Insight)** An observer looks at the top of a vertical tower from point A on a flat plain, noting an angle of elevation of 15°. They walk 100 meters in a straight line directly toward the base of the tower to point B, where the angle of elevation is now 30°. What is the height of the tower in meters?

6. **(Integer Bounds & Estimation)** In $\triangle ABC$, side $b = 20$ and $\angle A = 45^\circ$. According to the Law of Sines ambiguous case, exactly two distinct valid triangles exist if side a falls within a strictly bounded range. What is the sum

of the smallest and largest possible integer values for a that produce exactly two triangles?